

- **System:** Authenticate User, Recommend Package, Calculate Price, Process Payment.

#### 4.4.2 Sequence Diagram (Booking Process)

This diagram would illustrate the sequence of interactions between the User, App, API Gateway, Booking Service, and ML Service from the moment the user requests a quote to the final booking confirmation.

### 4.5 Core Feature Implementation (ML Models)

#### 4.1 Module 1: User Data & Inventory Management

- Dataset simulated with **10,000+ records**.
- Features include: total items, fragile items, distance, day of week, etc.

**Table 4: Sample Dataset Features**

| Feature            | Type        | Description                         |
|--------------------|-------------|-------------------------------------|
| total_items        | Integer     | Count of all items                  |
| fragile_item_count | Integer     | Count of fragile items              |
| distance_km        | Float       | Distance between source/destination |
| move_day_of_week   | Categorical | Day of week                         |
| move_hour          | Categorical | Hour of day                         |

#### 4.2 Module 2: ML-Powered Package Recommendation System